$\alpha: \underset{s}{I} \to \mathbb{R}^3$ p.a.l., $k>0$ $\qquad O_3 = \begin{pmatrix} 0&0&0\\0&0&0\\0&0&0 \end{pmatrix}$

$s \longmapsto \begin{pmatrix} T(s)\\ N(s)\\ B(s)\end{pmatrix} \in \mathbb{R}^9$ $\qquad Id_3 = \begin{pmatrix} 1&0&0\\0&1&0\\0&0&1\end{pmatrix}$

$$\begin{pmatrix} T'\\ N'\\ B'\end{pmatrix} = \begin{pmatrix} kN\\ -kT-\tau B\\ \tau N\end{pmatrix} = \begin{pmatrix} O_3 & kId_3 & O_3\\ -kId_3 & O_3 & -\tau Id_3\\ O_3 & \tau Id_3 & O_3\end{pmatrix}\begin{pmatrix} T\\ N\\ B\end{pmatrix}$$

Fundamental theorem of the local theory of curves:

Given $k_0, \tau_0 : I \to \mathbb{R}$, $\mathcal{C}^\infty$, $k_0>0$, there exists a curve $\alpha: I \to \mathbb{R}^3$ p.a.l. s.t. $k(s)=k_0(s)$ and $\tau(s)=\tau_0(s)$

Moreover, α is unique up to direct isometries of $\mathbb{R}^3$

pf: Look at $x'(s) = A_0(s)\,x(s)$ ····(*)

where $A_0(s) = \begin{pmatrix} O_3 & k_0(s)Id_3 & O_3\\ -k_0Id & O_3 & -\tau_0 Id_3\\ O_3 & \tau_0 Id_3 & O_3\end{pmatrix}$

Choose $\underline{a} \in \mathbb{R}^9$ s.t. $\underline{t}_0 = (a_1,a_2,a_3)$, $\underline{n}_0 = (a_4,a_5,a_6)$, $\underline{b}_0 = (a_7,a_8,a_9)$ form an oriented orthonormal basis of $\mathbb{R}^3$

Set $f: I \to \mathbb{R}^9$ solution of (*)

and define $\underline{t} = (f_1,f_2,f_3)$, $\underline{n} = (f_4,f_5,f_6)$, $\underline{b} = (f_7,f_8,f_9)$

We want to prove that $\{\underline{t},\underline{n},\underline{b}\}$ is an orthonormal (>0) basis of $\mathbb{R}^3$ $\forall s \in I$

look at $M(s) = \begin{pmatrix} \langle \underline{t},\underline{t}\rangle & \langle \underline{t},\underline{n}\rangle & \langle \underline{t},\underline{b}\rangle\\ \langle \underline{t},\underline{n}\rangle & \langle \underline{n},\underline{n}\rangle & \langle \underline{n},\underline{b}\rangle\\ \langle \underline{t},\underline{b}\rangle & \langle \underline{n},\underline{b}\rangle & \langle \underline{b},\underline{b}\rangle\end{pmatrix}$

$(*) \Rightarrow M'(s) = A(s)M(s) - M(s)A(s)$ where $A(s) = \begin{pmatrix} 0 & k_0 & 0\\ -k_0 & 0 & -\tau_0\\ 0 & \tau_0 & 0\end{pmatrix}(s)$, $M(0) = Id_3$

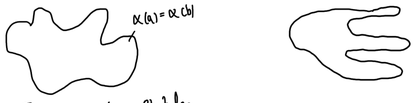
But $M(s) = Id_3$ $\forall s$ is a solution

$\alpha: I \to \mathbb{R}^3$

$s \to \int_{s_0}^{s} t(u)\,du$

Simple closed planar curves

$\alpha: \underset{[a,b]}{I} \to \mathbb{R}^2$ $\qquad \alpha(a) = \alpha(b)$, no self-intersection



Isoperimetric Problem

Among all planar simple closed curves of length L which one bounds the region of largest area?

Green's Thm: $f(x,y), g(x,y) \in \mathcal{C}^1(\mathbb{R}^2)$

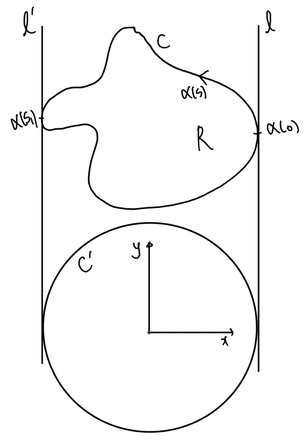
R is the bounded region inside a curve $C \in \mathbb{R}^2$

then, $\int_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) dx\,dy = \int_C \left(f\frac{dx}{dt} + g\frac{dy}{dt}\right) dt$

Take $f=-y$, $g=x$, $\quad 2\,\text{Area}(R) = \int_a^b \left(x(s)\frac{dy}{dt} - y(s)\frac{dx}{dt}\right) dt$

Theorem: $L^2 \geq 4\pi\,\text{Area}(R)$

Moreover, $L^2 = 4\pi\,\text{Area}(R)$ iff α parametrized a circle.



If C is parametrized by a.l. as $\alpha(s) = (x(s), y(s))$ $\qquad s \in [0,L]$

we can choose a parametrization of C' as $\beta(s) = (x(s), \bar{y}(s))$

$\text{Area}(R) = \frac{1}{2}\int_0^L (xy' - x'y)(s)\,ds = \int_0^L xy'\,ds$

$\left[\int_0^L x'y\,ds = (xy)\Big|_0^L - \int_0^L xy'\right]$

$\text{Area(interior of } C') = \pi r^2 = -\int_0^L x'\bar{y}\,ds$

$\text{Area}(R) + \pi r^2 = \int_0^L xy' - x'\bar{y}\,ds \leq \int_0^L \sqrt{(xy'-\bar{y}x')^2}\,dx \leq \int_0^L \sqrt{(x^2+\bar{y}^2)(x'^2+y'^2)}\,ds = \int_0^L \sqrt{x^2+\bar{y}^2}\,ds = Lr$

$\sqrt{\text{Area}(R)}\sqrt{\pi r^2} \leq \frac{1}{2}(\text{Area}(R) + \pi r^2) \leq \frac{Lr}{2}$ $\qquad$ "$=$" $\to$ exercise